Gate Synthesis for Arbitrary Z-Rotations Using Search-Based Methods

1 Background

1.1 Qubits

A qubit is the fundamental unit of quantum information. Unlike a classical bit, which exists strictly in one of two states (0 or 1), a qubit can exist in a superposition of both states:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \quad \text{where } |\alpha|^2 + |\beta|^2 = 1.$$

This enables a continuous state space represented geometrically on the Bloch sphere.

1.2 Quantum Gates

Quantum gates are unitary operations that transform qubit states. These transformations correspond to rotations on the Bloch sphere. For a single qubit, gates can be interpreted as rotations about axes in $\mathbb{R}^3$.

1.3 Clifford Gates

Clifford gates form a restricted but important subset of quantum operations. They map Pauli operators to Pauli operators under conjugation. Examples include:

- Hadamard (H)
- Phase (S)
- CNOT (multi-qubit)

These gates generate rotations constrained to discrete angular increments (multiples of $\pi/2$ or $\pi/4$), limiting expressivity.

1.4 T Gate

The T gate introduces a $\pi/8$ phase rotation:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

Unlike Clifford gates, the inclusion of the T gate enables universal quantum computation by allowing access to a denser set of rotations.

1.5 Z-Rotation Gates

The general rotation about the Z-axis is given by:

$$R_z(\theta) = e^{-i\theta Z/2}.$$

Clifford+T gate sets can exactly represent only a discrete subset of these rotations. Arbitrary θ must be approximated.

1.6 A* Algorithm

A* is a search algorithm that finds optimal paths by minimizing:

$$f(n) = g(n) + h(n),$$

where:

- $g(n)$: cost from start to node n
- $h(n)$: heuristic estimate from n to goal

It combines properties of breadth-first search and greedy best-first search.

2 Motivation

Synthesizing arbitrary $R_z(\theta)$ rotations is straightforward for special angles such as:

$$R_z(\pi/2) = S, \quad R_z(\pi/4) = T.$$

However, for angles such as $\pi/8$, no exact finite representation exists using Clifford+T gates. Instead, one must construct sequences that approximate the desired rotation.

3 Geometric Interpretation

Quantum gate sequences correspond to trajectories on the Bloch sphere.

The figure shows that achieving higher precision requires exponentially longer gate sequences.

4 Limitations of Clifford Gates

Sequences such as HS or HT produce rotations confined to discrete angular grids (multiples of $\pi/4$ or $\pi/2$).

This discretization prevents representation of arbitrary angles.

5 Non-Clifford Behavior

Introducing T gates leads to more complex and less predictable trajectories.

Unlike pure Clifford sequences, compositions including T do not necessarily return to structured paths.

These sequences can approximate arbitrary rotations but are difficult to predict analytically.

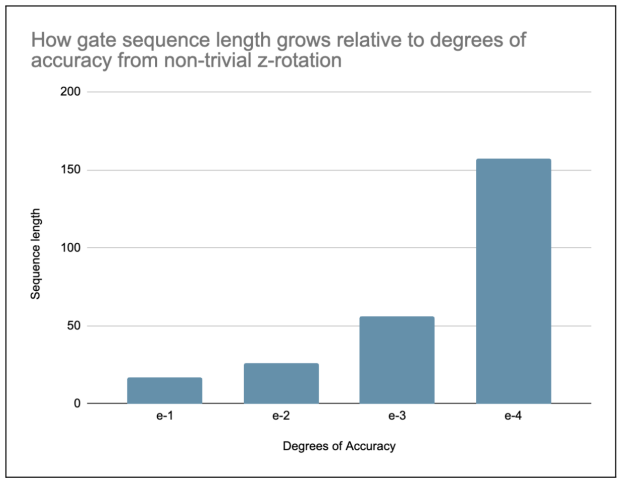

Figure 1: Growth in gate sequence length as approximation accuracy increases

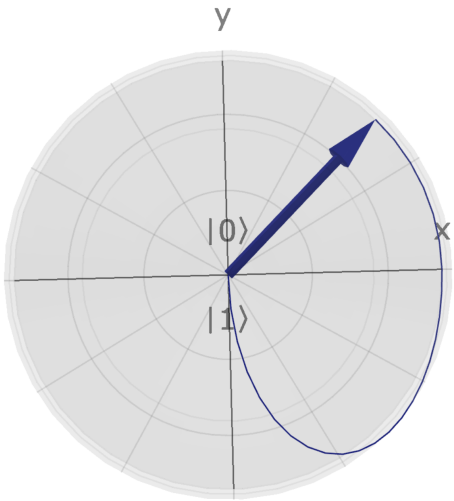

Figure 2: H Gate Followed by T gate

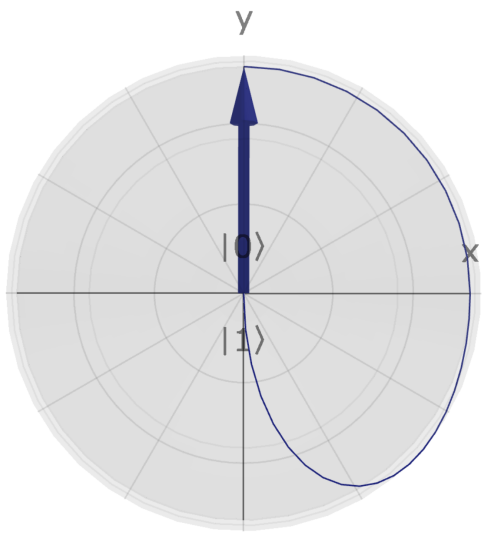

Figure 3: H followed by S. Clifford gate rotations constrained to discrete angles

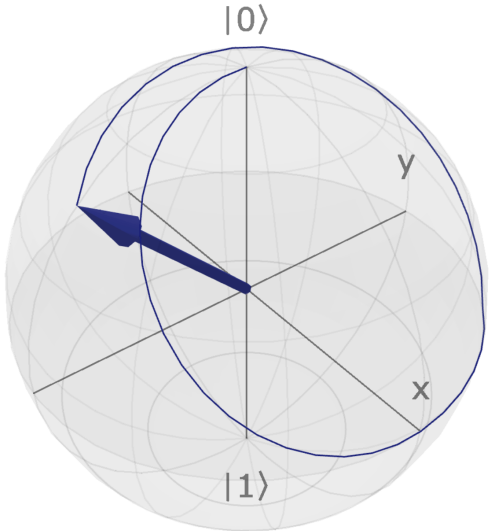

Figure 4: Effect of HTH on trajectory being no longer bound by the basic angular grid.

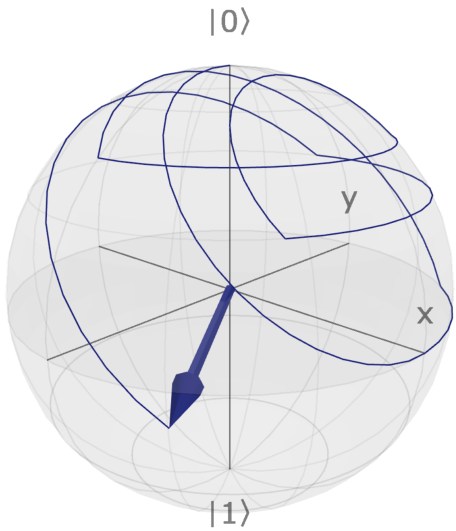

Figure 5: Example sequence $HTHSTHSTH$ producing complex trajectory that is close enough to the XY plane that it can be used to approximate an angle on it.

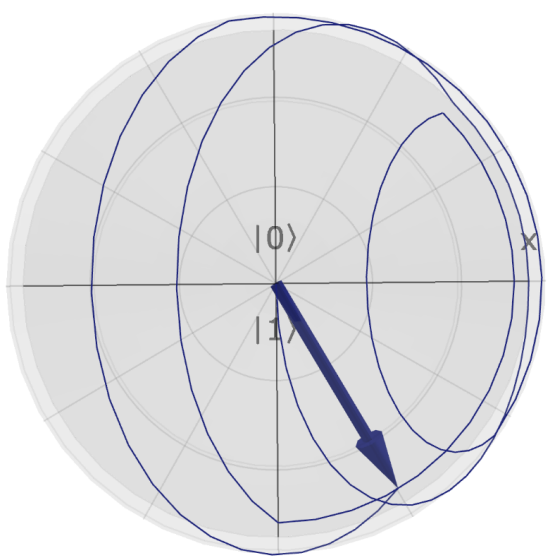

Figure 6: The same view from above. See that the angle is now closer to a 22.5° subdivision.

6 Search-Based Synthesis

6.1 Brute Force Approach

A naive approach is to enumerate gate sequences using:

- Breadth-first search (BFS)
- Depth-first search (DFS)

However, several issues arise:

- Redundant sequences (e.g., $HH = I$, $SSSS = I$)
- Cycles and loops
- Equivalent transformations (e.g., gate identities)

These require pruning rules such as:

- Avoid consecutive identical gates
- Eliminate inverse pairs
- Detect equivalent unitary operations

Despite pruning, complexity grows rapidly.

6.2 A*-Based Approach

To improve efficiency, we model synthesis as a search problem.

We define:

$$f = s + w \cdot d,$$

where:

- s: cost from start (sequence length)
- d: distance from current state to target rotation
- w: weighting factor

The algorithm proceeds by:

1. Expanding candidate sequences
2. Evaluating their Bloch sphere positions
3. Selecting the next expansion based on minimal f

This prioritizes sequences that both:

- Remain short
- Approach the target angle

7 Results

As shown earlier, increased accuracy demands longer gate sequences. The A*-based approach balances accuracy and cost, producing near-optimal approximations of rotations such as $R_z(\pi/8)$.

8 Team synthesis part 3

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

Now, we extend and also add a restriction onto our system. We now have two qubits instead of 1, have access to CNOT, but we CANNOT use T on the main qubit. So, we need to use quantum teleportation of the T gate onto the qubit.

Here's the setup:

Imagine your main is in some state

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

And your ancilla qubit is in $|0\rangle$.

Now, if you *were* able to apply the T-gate onto your state $|\psi\rangle$, you'd get the state $\alpha |0\rangle + e^{i\pi/4}\beta |1\rangle$. Notably, the problem restricts this. How do we get around it?

To the ancilla bit, starting from $|0\rangle$, apply H then T to put the state into the $1/\sqrt{2}(|0\rangle + \beta e^{i\pi/4} |1\rangle)$. From here, apply the $CNOT$ gate, where the *main* qubit is the control, and the *ancilla* is the target. This forms the following state:

$|\psi\rangle \otimes |T\rangle = 1/\sqrt{2}(\alpha |0\rangle |0\rangle + \alpha e^{i\pi/4} |0\rangle |1\rangle + \beta |1\rangle |0\rangle + \beta e^{i\pi/4} |1\rangle |1\rangle)$.

But wait, there is one piece of information missing.

T is not a high-fidelity gate. T requires a conecpt called "distillation", where the following happens. 15 physical qubits (in $|0\rangle$) have the H applied to all of them (i.e they're now all in $|+\rangle$). Then, the noisy T gate is applied to all of them. Then, they're entangled with a mesh of CNOTs based on the [15,1,3] Reed-Muller code, in which 1 physical qubit is set as the surivor/clean node, and the rest are "used as checks". This process results in a high-fidelity T but at the expense of extra setup. This is the dominant cost of having T gates, and this issue occurs with each time we setup the $|T\rangle$ state, i.e $T|+\rangle$

From here, do a Z-basis measurement, which will return a bit m.

If $m = 0$, then the T gate has been applied, no worries. If $m = 1$, however, then $T\dagger$ has been applied, meaning that we need to apply a S gate to move the state into T from $T^\dagger$, then *rebuild* the whole subroutine (using a reset)

How do I know it's been applied?

Well, if we go from the CNOT state that I had placed above, $1/\sqrt{2}(\alpha |0\rangle |0\rangle + \alpha e^{i\pi/4} |0\rangle |1\rangle + \beta |1\rangle |0\rangle + \beta e^{i\pi/4} |1\rangle |1\rangle)$, if m, the second value, is 0, then you keep the $\alpha |0\rangle + \beta e^{i\pi/4} |1\rangle$, as those are the two that had the second qubit as 0. This forms $\alpha |0\rangle + \beta e^{i\pi/4} |1\rangle$, which is equivalent to applying T onto the original $|\psi\rangle$. If the measurement result was $m = 1$, we'd need to take the states where the second qubit is 1, meaning $\alpha e^{i\pi/4} |0\rangle + \beta |1\rangle$, which is equivalent to applying $T^\dagger$ onto the original $|\psi\rangle$

This, therefore, incurs a cost onto the sequences we form from the previous section (that we got from A*). For every T, we need to apply a Hadamard, a T gate, a CNOT, and a feed forward. Moreovoer, the measurement bit that decides application m will be $m = 1$, i.e not correct and requiring one extra S gate to go back to T ($ST^\dagger = T$) (on top of rebuilding the previous), 50

Team synthesis part 4 To get started, the Steane code maps 7 physical qubits to 1 physical qubits. These 7 serve the following purposes: - 3 qubits: Phase flip error - 3 qubits: Bit flip error - 1 qubit: Holds the actual information

The way this is implemented is via 3 Hadamards and a mesh of CNOTS. We can construct it with a kernel, which we'll do below

Now, for logical qubits, the entire Clifford set is transversal, meaning the logical H, S, CX are those gates applied to all the physical qubits. Notably, the T gate is *not transversal*. We need to form a logical $|T\rangle_l$ gate.

The methodology to form this is found inside of "Experimental demonstration of logical magic state distillation" paper by QuEra, specifically figure 2A for forming an arbitrary state.

In the bottom, the arbitrary physical state is passed in. For us, this is the $|T\rangle$, which is formed as T acting on $|+\rangle$.

Notably, the paper uses $\sqrt(Y)$ and $\sqrt(Y)^\dagger$, due to hardware efficiency. Our best bet is to try and map this to our set of $H, S, CNOT$ in order to form $|T\rangle_{logical}$. Following this idea, this means that for every T that shows up in our process that we want to teleport to the logical state, we need to reform this entire setup, make $|T\rangle_{logical}$ from $|T\rangle_{physical}$, then teleport the logical T to the logical state, then continue. So, the subroutine has gotten substantially more complicated, as not only do you require the T distillation (the T state at the beginning passed in at the bottom), but you also need to do this entire circuit to have a logical T gate.

With this in mind, we can define a subroutine that uses 6 ancilla qubits, with 1 qubit at the end representing the physical $|T\rangle$ state that follows with the QuEra paper.

After forming the logical T, it can get teleported similarly to part 3, where it gets observed in the Z-basis, and we feed forward the signal from the measurement: if m =0, do nothing, if m = 1 then do a transversal S. This is ok to do because the transversal S is relatively cheap.

Notice that forming the T gate (at the beginning of the logical T), requires an entire 15-qubit distillation process (as I mentioned in part 3), AND this extra encoding to make it $|T\rangle_{logical}$, meaning that applying a T gate is a very expensive operation, despite just being 1 qubit.

Compare that to a $CNOT$, a two qubit gate, which doesn't exhibit these pitfalls. So, indeed, Hamlet and Qoratio, 1 qubit *can* be harder than a 2 qubit gate.

For AI usage, I used AI for topic clarification, specifically to elaborate what information was shown in the tutorial for teleportation of the gate from an ancilla qubit.